3D Soil & Waste Pipes

User guide · IngeTrazo plugin · `soil_waste_pipes.py` · Ctrl+Alt+W

Builds full-size soil stacks and outside sink waste pipes, with their fittings, each in its own group. Every new part joins onto the open end of the last, and the window stays open while you work, so you can build a whole run part by part.

Installing and updating

1. Open File Explorer, paste `C:\Users\eddy_\AppData\Roaming\ingetrazo\plugins` into the address bar and press Enter.
2. Copy `soil_waste_pipes.py` into that folder. When updating, choose **Replace the file**.
3. Restart IngeTrazo. Settings are remembered only while IngeTrazo is open, so a restart also clears old settings.

Keep only one copy of the plugin in the folder. Two copies under different names may both load.

Shortcut: Ctrl+Alt+W opens the window, or brings it to the front if it is already open.

From the Python console: paste the file's code in, select what the plugin needs, then type `run()`. If it can't find the 3D view by itself, use `run(viewport)`.

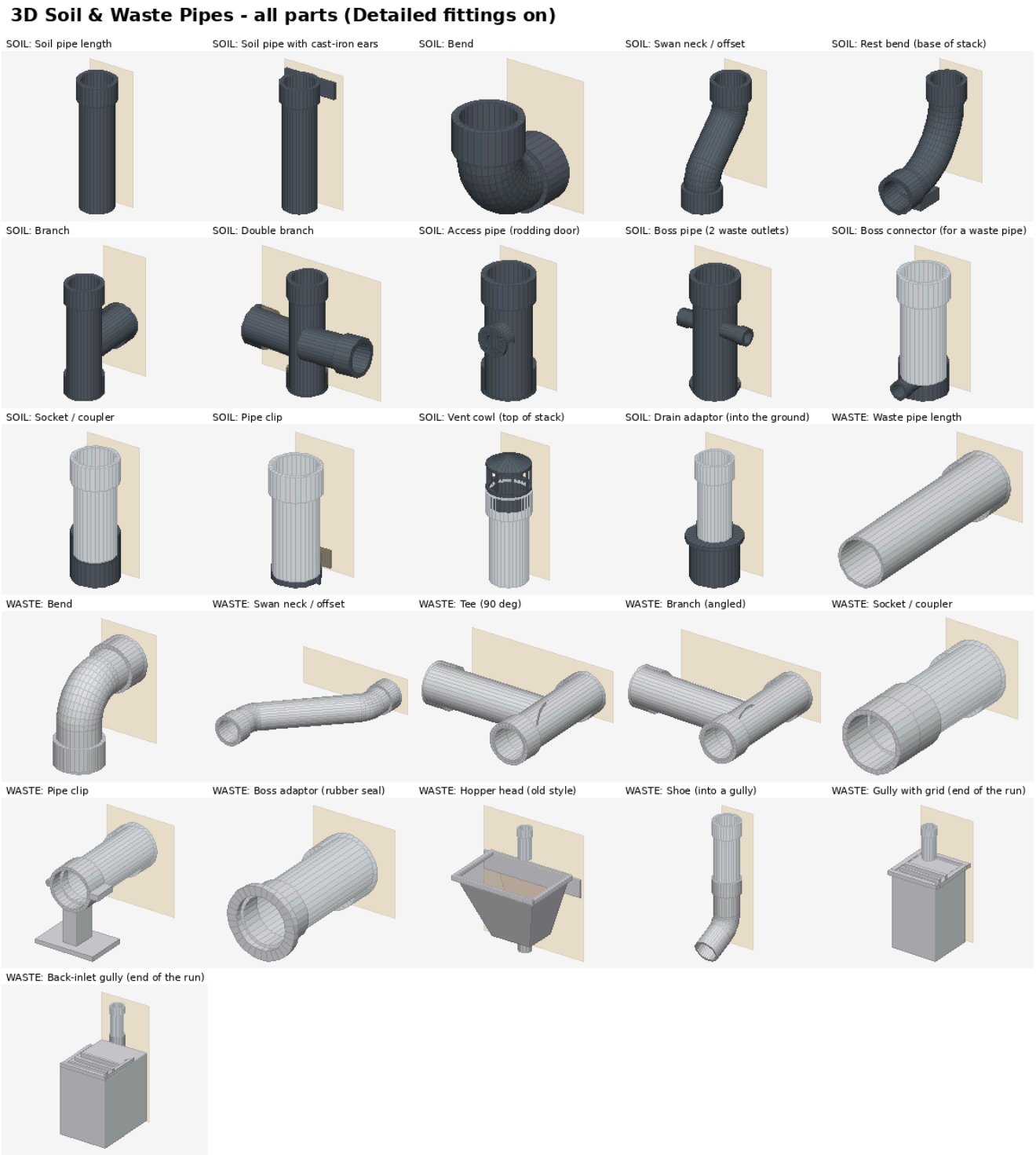
The window

Area	What it does
Top line	Says where the next part will go, for example "joins onto the far end of: Soil pipe length"
Solid (for 3D printing)	Fills the pipes in, with no hollow inside
Detailed fittings	Thick socket collars on fitting ends, a round rodding eye with cap and bolts, collars on boss outlets, a cage vent, a stepped drain adaptor and hinged clips
Pipe lengths have a socket	Puts a socket collar on the start (top) end of every pipe length, like real pipe
Cast-iron style ears	Two fixing lugs on each pipe socket, facing the wall, instead of separate clips
Preset	Picks a saved set of sizes; **Save as**, **Delete** and **Open folder** manage your own presets
Part list	Soil pipe parts and waste pipe parts
Pipe sizes	Soil and waste pipe diameters and wall thicknesses, in mm
This part	A live 3D preview, plus the settings the chosen part uses
Start on a wall	Starts a soil stack down a wall, or a waste pipe out of it
Pipe clips	Automatic clips along pipe lengths, and chunky fittings for printing
Buttons	**Add part**, **Use selected end**, **Start at the origin**, **Close**

In the preview, soil pipe is dark grey, waste pipe is white, the brown panel is the wall, and the pale grey pipe is the one a fitting sits over.

All sizes are typed in millimetres. IngeTrazo stores geometry in metres whatever your display units, and the plugin converts for you.

Parts



System	Part	Settings it uses	Notes
Soil	Soil pipe length	Length	Socket on the top end; gets clips, or ears if ticked

System	Part	Settings it uses	Notes
Soil	Bend	Turn angle, direction	Turn 87.5 = 92.5° bend, 67.5 = 112.5°, 45 = 135°
Soil	Swan neck / offset	Turn angle, direction, offset length	Two bends with a straight between; steps the stack sideways
Soil	Rest bend	Direction	Long-radius bend with a foot, for the base of a stack
Soil	Branch	Turn angle, direction, branch length	The arm angles back up towards the incoming flow
Soil	Double branch	Turn angle, direction, branch length	Two arms on opposite sides; both ends can be joined onto
Soil	Access pipe	Direction	Round rodding eye with screw cap and bolts on the side the direction points to
Soil	Boss pipe	Direction	Two socketed waste outlets on opposite sides, and a base ring
Soil	Boss connector	Direction, distance back from the end	Strap boss on the stack; its outlet takes a waste pipe
Soil	Socket / coupler	Distance back from the end	Two collars back to back over a joint
Soil	Pipe clip	Direction, distance back from the end	Hinged clip with a fixing lug and wall plate
Soil	Vent cowl	—	Cage cowl on the top of the stack; closes the run
Soil	Drain adaptor	—	Stepped adaptor into the underground drain; closes the run
Waste	Waste pipe length	Length	Socket on the start end; gets clips, or ears if ticked
Waste	Bend	Turn angle, direction	Use a turn of 90 for a 90° bend
Waste	Swan neck / offset	Turn angle, direction, offset length	Two bends with a straight between
Waste	Tee (90°)	Direction, branch length	Branch at right angles
Waste	Branch (angled)	Turn angle, direction, branch length	Turn 45 = 135° branch
Waste	Socket / coupler	Distance back from the end	Two collars back to back over a joint
Waste	Pipe clip	Direction, distance back from the end	Hinged clip with a fixing lug and wall plate
Waste	Boss adaptor (rubber seal)	Distance back from the end	Sleeve with a lip over a boss outlet or waste end
Waste	Hopper head (old style)	—	Open-topped hopper on the wall; waste pipes discharge into it, the downpipe carries on from its outlet
Waste	Shoe	Direction	Kicks the end of a pipe out over a gully; closes the run
Waste	Gully with grid	—	Sits 30 mm under a downward pipe end; closes the run
Waste	Back-inlet gully	—	The pipe goes down into a socket at the back, below the grid; closes the run

With **Detailed fittings** unticked, every part is built in its simple shape.

Direction sets which way a bend, branch, boss outlet, access door or clip faces. On a pipe coming out of a wall, **Down** bends it down. On an upright pipe, the same setting turns it towards the wall.

Building a run

1. Choose a starting point. With nothing chosen, the first part goes at the origin: soil pipe runs down, waste pipe comes straight out of the wall. **New run - wall is** sets which side the wall is on. To start on a building, see **Starting on a wall** below.
2. Pick a part, check it in the preview, and click **Add part**.
3. The next part joins onto the far end of the one you just added. Keep adding parts.
4. To join a waste pipe into a stack, add a **Boss connector** on the soil pipe, then choose a waste part. It goes onto the boss outlet automatically.
5. A vent cowl, drain adaptor, shoe or gully finishes a run. The window then moves to the newest end still open, for example the top of the stack for the vent.
6. For an old-style hopper, add the **Hopper head** (on a wall or under a waste pipe end), then carry on down from its outlet with waste pipe lengths. Set the waste diameter to your downpipe size, for example 68 mm, for the pipe below it.

To join onto any other end, click that open end face in the model, then **Use selected end**. You can move or rotate placed parts with IngeTrazo's own tools and still join onto them. If you undo a part, click an end and **Use selected end**, because the window still points at the removed part.

Starting on a wall

Select a wall face and click **Use selected wall**. The next part starts on that wall:

- **A soil part** starts a stack at the top of the wall and runs down to the bottom, with its clips flat on the wall.
- **A waste part** comes straight out of the wall at the height you set. Add a **Down** bend, a waste pipe and a gully to finish it.
- **A hopper head** stands on the wall with its back plate flat on it and its rim at the waste height.

Setting	Default	Effect
Along the wall	500 mm	Distance from the left end of the wall, as you look at it from outside
Soil - below top of wall	0 mm	Where the stack starts. Negative takes it above the wall, for example to vent above the eaves
Soil - below bottom of wall	0 mm	How far the stack carries on into the ground
Waste - height on the wall	750 mm	Height above the bottom of the wall where the waste comes out
Extra gap from wall	0 mm	0 puts the soil clips flat on the wall
Soil pipe length = wall height	On	The first soil pipe length fits the wall, including the two soil settings
Wrong side?	Off	Tick it if the pipe appears on the inside of the wall

Pipe clips

With **Add them evenly** ticked, each pipe length gets clips, with their lugs pointing at the wall. With **Cast-iron style ears** ticked, pipe lengths get ears on their sockets instead, and no clips are added. One goes about 150 mm from each end, and the rest are spaced evenly between, no further apart than you set: 1800 mm for soil and 500 mm for waste by default. They go in their own group, and a single undo removes them together with their length.

Chunky fittings (for 3D printing) makes every clip, socket, boss and vent thicker, 3× by default, and closes the gap between each fitting and its pipe so they print as one piece. For small-scale printing, tick both **Solid** and **Chunky fittings**, then scale a copy with 3D Print Scaler.

Presets

Four presets are built in:

- UK 110 soil + 40 waste
- UK 110 soil + 32 waste
- UK 110 soil + 50 waste
- UK 82 soil + 50 waste

Save as stores your current sizes and options as one `.json` file in Documents › IngeTrazo Presets › Soil and Waste. **Open folder** opens that folder, so you can copy or share preset files.

Troubleshooting

Problem	Cause	Fix
"A waste pipe part can't join onto a soil pipe end"	The two systems only join through a boss	Add a **Boss connector** on the soil pipe first
A new part doesn't join where expected	The window still points at an undone or deleted part	Click the right open end and use **Use selected end**
The pipe appears inside the wall	The wall face points inwards	Tick **Wrong side?** in Start on a wall
The gully, hopper or shoe is tilted	It was added to a pipe that isn't pointing down	Add a **Down** bend and a short pipe first
"Along the wall is past the right-hand end"	The setting is longer than the wall	Make **Along the wall** smaller
The shortcut is missing in the extensions window	Another command already uses Ctrl+Alt+W	Check Preferences › Keyboard shortcuts

Limits

- Boss outlets and branches don't cut a hole into the main pipe.
- Gullies and the hopper are simple models: bodies with a grid or an open top.
- Joints are butt joints, not overlapping sockets.
- Joining works only on ends made by this plugin.